Power Budget

Team Number:	210
Project Name:	**VibeWater**
Team Member Names:	**Asadbek, Neel, K, Jacob**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Opamp	MCP6004	+1.8 to 6V	1	30	30	mA
	5V regulator	LM7805	+5V - 35V	1	1000	1000	mA
	PIC Discovery Nano	PIC18F577Q43	1.8-5.1v	1	500	200	mA

+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	PIC Discovery Nano	PIC18F577Q43	1.8-5.1v	1	500	500	mA
	Speaker Drive Circuit	Customer	1.8-5.1v	1	30	30	mA
						0	mA
						0	mA
						0	mA
					Subtotal	530	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	662.5	mA
c2. Regulator or Source Ch	+5V Regulator	LM7805	(range)	1	1000	1000	mA
					Total Remaining Current Available on +5V Rail	337.5	mA

+3.3V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	LCD screen	NHD-0216K1Z-NSW-BBW-L	2.8-3.1V	1	10mA	10	mA
						0	mA
						0	mA
						0	mA
					Subtotal	10	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	12.5	mA
c4. Regulator or Source Ch	+3.3V low-dropout regulator	KA78RM33RTF	+5V - 20V	1	500	500	mA
					Total Remaining Current Available on 3.3V Rail	487.5	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each

External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	SWI12-12-E-P5 9V 2A AC DC Power	110VAC	+9V	5000	5000	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	(range)	1	1000	1000	mA
					Total Remaining Current Available on External Power Source 1	4000	mA